

Calibrated Measurement of Speaker Identity and Prosody Transfer in Cross-Lingual Speech Synthesis

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Abstract—Zero-shot cross-lingual voice cloning is routinely described as preserving speaker identity, and the claim is supported by a cosine similarity between speaker embeddings. Such a score has no natural scale, because no reference exists for how similar the same person sounds to themselves after switching language. We remove that limitation by evaluating on a same-speaker parallel corpus, in which one bilingual speaker re-enacts their own utterances in the other language, so that every synthesised utterance has a matching human recording of the same speaker performing the same content. Using a cascade of Whisper, MarianMT and XTTS-v2 evaluated on 870 held-out English–Spanish utterances over a speaker-disjoint partition, we report two calibrated results. Speaker identity reaches 94.5 % (EN→ES) and 86.1 % (ES→EN) of the human cross-language ceiling, which is itself only 0.445 under an ECAPA cosine. Prosody does not follow: within a speaker, human cross-lingual pitch level transfers at $r = 0.849$, whereas synthesised speech reaches 0.311 and 0.314, and pitch range collapses from 0.214 to 0.035–0.040. We show that pooled correlations overstate transfer roughly twofold because they absorb stable between-speaker variation carried by the voice clone, and that the predictable component of cross-lingual prosody lies in pitch range rather than pitch level or timing: a ridge model beats a copy-plus-offset baseline by 5.1 % and 3.9 % mean absolute error for pitch range under family-wise correction, with a negative learned coefficient indicating compression towards the speaker’s habitual range. We also report a measurement pitfall that inverted an earlier version of this result, and give the corrected procedure.

Index Terms—Speech-to-speech translation, speaker identity, cross-lingual prosody, voice cloning, speech synthesis evaluation, automatic dubbing.

I. INTRODUCTION

SPEECH carries what was said and how it was said. Cross-lingual speech synthesis has become good at transferring the first, and zero-shot voice cloning [1]–[3] is widely reported to transfer a large part of the second, in the sense that the output is recognisably the same speaker. The supporting evidence is normally a cosine similarity between speaker-verification embeddings of the source recording and the synthesised output.

That evidence has an unresolved scale problem. A cosine of 0.42 is neither good nor bad on its own, because human speakers do not sound identical to themselves across languages either: phonetic inventory, rhythm class and habitual pitch

range all shift. Part of the apparent loss belongs to the language change rather than to the synthesiser, and no standard evaluation separates the two.

A second problem is that identity and prosody are reported together under headings such as naturalness, although they have different mechanisms. A voice-cloning synthesiser is conditioned on a reference recording for *timbre*. It is not given the source utterance’s pitch contour, so it must infer intonation from the target text. Whether the resulting prosody resembles what the speaker would actually have produced in the target language is an empirical question that can only be answered if a recording of that speaker doing exactly that exists.

We answer both questions by evaluating on the Dialogs Re-enacted Across Languages (DRAL) corpus [4], in which bilingual participants hold a spontaneous conversation and then re-enact each fragment in the other language. Both sides of every pair come from one speaker performing one communicative act twice. This supplies, per utterance, a *human cross-language anchor* for identity and a *human cross-language reference* for prosody.

The contributions of this paper are:

- 1) An identity result expressed as a percentage of a same-speaker cross-language ceiling rather than as an unanchored cosine, over 870 held-out utterances in both directions.
- 2) A quantification of the cross-lingual prosody transfer gap, reported within-speaker, with a demonstration that pooled correlations overstate transfer roughly twofold by absorbing the between-speaker variation that voice cloning already carries.
- 3) Evidence that the predictable component of cross-lingual prosody lies in pitch range rather than pitch level or timing, with a learned coefficient that supplies the mechanism.
- 4) A measurement pitfall, quantified: correlating fundamental frequency in hertz without a plausibility gate understates the correspondence by roughly a factor of seven, and the intuitive remedy of narrowing the pitch tracker’s search band makes it worse.

II. RELATED WORK

A. Cascaded and direct speech-to-speech translation

Direct models map source speech to target speech through spectrograms [5] or discrete units [6], [7], avoiding error propagation and in principle carrying paralinguistic information that text discards. Cascades remain competitive in quality [8] and, decisively for a study whose object is measurement, are interpretable stage by stage [9].

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Code, result files and the scripts that regenerate every number in this paper are available in the project repository.

B. Identity-preserving synthesis

Zero-shot multi-speaker synthesis conditions the decoder on an embedding from a separately trained speaker-verification network [1], extended to the multilingual case by YourTTS [2] and scaled by XTTS-v2 [3]. The embedding is trained for verification and is therefore designed to be invariant to what a speaker does on any particular utterance — which is precisely what prosody consists of.

C. Cross-lingual prosody transfer

Within a language, prosody transfer from a reference recording is comparatively mature [10], [11]. Across languages it is limited by data: parallel expressive recordings of the same content in two languages are rare. Swiatkowski *et al.* [12] learn a prosody representation transferable across language and speaker for machine dubbing, using a variational reference encoder with noise modelling for field recordings — a design made necessary by the absence of clean parallel data. StyleS2ST [13] pursues zero-shot style transfer inside a direct model, MSLM [14] preserves speaker style in a textless speech language model, and Seamless [15] adds expressivity to large-scale multilingual translation. Our contribution is complementary: not a better transfer mechanism, but a calibrated statement of the size of the gap those mechanisms exist to close.

D. Evaluation practice

Three weaknesses recur. Speaker-similarity cosines are reported without a same-speaker cross-language ceiling. Intrusive signal metrics such as PESQ and STOI are proposed although they require a time-aligned reference of the same utterance, which cannot exist across languages. And correlations are reported pooled across speakers, counting stable between-speaker differences as evidence of transfer. We address all three explicitly.

III. METHOD

A. Corpus and partitioning

The 16 kHz public-domain release of DRAL [4] supplies 2,893 linked English–Spanish fragment pairs over 96 speaker identifiers, 4.27 hours of speech, of which 2,834 pairs are performed by a single speaker. We partition 70/15/15 with seed 498. Because 59 pairs link fragments whose two sides carry different speaker identifiers, assignment operates on connected components of the speaker-identity graph, so that no speaker, and no linked group of speakers, appears on both sides of the train/test boundary. The test split holds 435 pairs and 14 speakers.

B. System

The system is a cascade: Whisper `small` [16] transcribes with the source language supplied explicitly; MarianMT [17] in the OPUS-MT bilingual configurations [18] translates; and XTTS-v2 [3] synthesises in the target language conditioned on the *source* recording as voice reference. The synthesis

seed is derived deterministically from the pair identifier and direction, so any utterance can be regenerated exactly. Failures are written to the results file with their failing stage and are counted in the reported rate rather than discarded.

C. Identity against a human ceiling

Let $e(\cdot)$ denote an ECAPA-TDNN embedding [19] computed at 16 kHz mono. For every utterance we compute

$$S_{\text{sys}} = \cos(e(x_{\text{src}}), e(\hat{y})), \quad (1)$$

$$S_{\text{human}} = \cos(e(x_{\text{src}}), e(y_{\text{human}})), \quad (2)$$

where x_{src} is the source recording, $\hat{y}$ the synthesised output and y_{human} the *same speaker's* real recording in the target language. We report $100 S_{\text{sys}}/S_{\text{human}}$. Both quantities are averaged within each speaker before averaging across speakers, because the test split ranges from 2 to 118 pairs per speaker.

D. Prosodic features and two frequency bands

We extract fundamental frequency with pYIN [20] in librosa [21], frame-level RMS energy, duration, and speaking rate in words per second. Two bands are used and must be kept distinct.

The *search band*, 65–1000 Hz, is given to the estimator. It must be wide: pYIN's voiced/unvoiced decision depends on the number of frequency candidates, so a narrow band lowers the aggregate voiced probability until voicing is not detected at all on quiet recordings. The *plausibility band*, 60–400 Hz, is applied downstream as a data-quality gate, since a median F0 outside the adult speaking range is a tracking failure rather than a voice. All F0 statistics are computed in semitones, $s = 12 \log_2(f/100)$, because pitch perception is logarithmic. Section V-A quantifies what happens when these choices are made incorrectly.

E. Within- and between-speaker decomposition

A correlation pooled over utterances from many speakers conflates two effects: the tendency of a low-pitched person to remain low-pitched in either language, and the transfer of utterance-level variation. Only the second is prosody transfer. We therefore report, for every measure, the pooled correlation, the between-speaker correlation over one point per speaker, and the *within-speaker* correlation obtained after centring both variables inside each speaker. The within-speaker figure answers the question.

F. Prosody prediction

To test whether target-language prosody is predictable at all, we predict target F0 level, target F0 range and $\log(\text{target duration}/\text{source duration})$ from six source-side features (F0 level, F0 range, log duration, voiced ratio, word count, speaking rate) against three baselines fitted on the training split: B0 copies the source value; B1 predicts the training mean; B2 adds a single global offset to B0. The model is closed-form ridge regression [22] fitted to the *residual* from B0, which makes it strictly nest B2, so it can only improve on

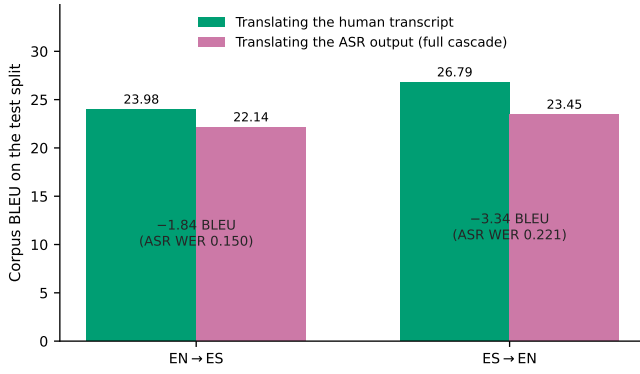


Fig. 1. Cost of automatic recognition in the cascade. Same translation model, same test utterances, identical decoding; the only difference is whether the input is the human transcript or the recogniser output.

the baseline up to estimation noise. The regularisation strength is chosen on the development split by the one-standard-error rule; the test split is scored once.

Because the test split holds 14 speakers and errors are correlated within a speaker, significance is assessed by a Wilcoxon signed-rank test [23] on per-speaker mean absolute errors and by a bootstrap over speakers [24]. Six models are fitted, so the family-wise position is reported using Bonferroni and Benjamini–Hochberg [25].

A quality gate is applied before analysis: both sides of a pair must yield an F0 estimate, a voiced-frame ratio of at least 0.30, a duration between 0.4 and 20 s, and a mean F0 inside the plausibility band. Restricted further to single-speaker pairs, 2,304 of 2,893 pairs remain.

IV. RESULTS

A. Reliability and intelligibility

Of 870 attempted utterances, 868 completed (failure rate 0.23 %); both failures were reference clips shorter than the synthesiser’s 0.33 s minimum, namely the backchannels “*Mm-hmm*.” and “*Yeah*.” Re-transcribing the synthesised audio and comparing against the text sent to the synthesiser gives a corpus word error rate of 0.081 (EN→ES) and 0.087 (ES→EN), with three empty transcriptions out of 868. The synthesiser produces clearly articulated speech; whatever it fails to preserve, it is not intelligibility.

Recognition error is 0.150 on English source audio and 0.221 on Spanish. Comparing the same translation model on the human transcript against the same model inside the cascade isolates the cost of recognition: 1.84 BLEU (EN→ES) and 3.34 BLEU (ES→EN), the larger loss occurring in the direction where recognition is worse (Fig. 1).

B. Identity against the human ceiling

Speaker similarity between source and synthesised output is 0.421 (EN→ES) and 0.383 (ES→EN). The human cross-language anchor is 0.445 in both directions, so the system retains **94.5 %** and **86.1 %** of the achievable similarity (Fig. 2).

Three points follow. The ceiling itself is low: two genuine recordings of one person in two languages are only 0.445

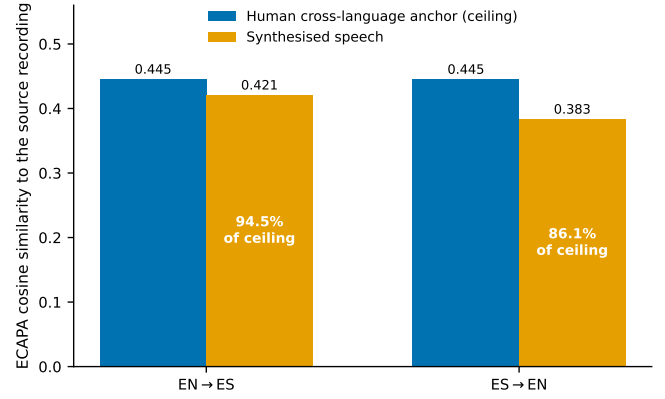


Fig. 2. Speaker identity against the human cross-language ceiling. The ceiling is the ECAPA cosine between the source recording and the *same speaker’s* real recording in the other language.

TABLE I
CROSS-LINGUAL PROSODY CORRESPONDENCE WITH THE HUMAN
TARGET-LANGUAGE RECORDING. WITHIN-SPEAKER REMOVES EACH
SPEAKER’S OWN MEAN BEFORE CORRELATING.

Measure	Human EN↔ES		System EN→ES		System ES→EN	
	Pool.	With.	Pool.	With.	Pool.	With.
F0 level	0.952	0.849	0.698	0.311	0.754	0.314
F0 range	0.263	0.214	0.142	0.035	0.132	0.040
Duration	0.870	0.867	0.364	0.352	0.618	0.610
Speaking rate	0.595	0.601	0.360	0.372	0.353	0.346
Energy	—	—	0.076	0.090	0.057	0.060
<i>n</i>	2,304		345–433		374–435	

similar, so an evaluation that treats a raw cosine of 0.42 as poor is applying a standard human speech does not meet. The two ceilings are identical to three decimal places, as they must be, since each compares the same pair of human recordings with the roles exchanged; the asymmetry between 94.5 % and 86.1 % is therefore entirely in the synthesiser’s output, and cloning works better into Spanish than into English. Finally, per speaker the percentage of ceiling ranges from 74.7 % to 147.2 % (EN→ES) and 55.5 % to 107.9 % (ES→EN); values above 100 % arise where the synthesised audio inherits channel characteristics from the reference clip that the human’s second recording does not share.

C. Cross-lingual prosody: human and system

Table I reports the correspondence between each measure and the human target-language recording, for the human re-enactment and for the system, pooled and within-speaker. Fig. 3 plots the within-speaker columns.

A human reproduces their own pitch level across languages at $r = 0.849$ within-speaker; the system reaches 0.311 and 0.314, about 37 % of the human figure. Pitch range collapses from an already modest human 0.214 to 0.035 and 0.040: the utterance-level variation that distinguishes an animated turn from a flat one does not survive. Duration survives best (0.352 and 0.610 against 0.867), which is expected since utterance length is largely determined by content. Energy carries es-

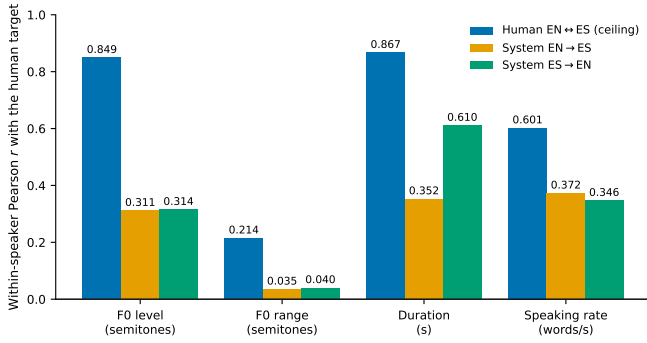


Fig. 3. Within-speaker cross-lingual prosody correspondence. Blue: what a human achieves re-enacting their own utterance in the other language. Orange and green: what the system achieves against that same human recording.

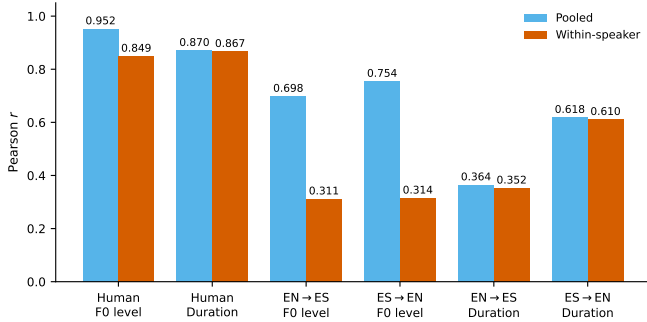


Fig. 4. Pooled against within-speaker correlation. Where the two diverge, most of the pooled figure is stable between-speaker variation carried by voice cloning rather than prosody transfer.

entially nothing, as expected for a quantity dominated by recording gain.

D. Why pooled correlations must not be quoted alone

For the system’s pitch level the pooled correlation is 0.698 and 0.754, while within-speaker it is 0.311 and 0.314; the between-speaker correlation is 0.895 and 0.847, and speaker differences account for 47% and 61% of total variance (Fig. 4).

A pooled correlation near 0.75 would ordinarily be reported as strong prosody transfer. It is not: it is the voice clone reproducing the speaker’s habitual pitch, which is the identity result of Section IV-B appearing again under a different name. Reporting the pooled figure alone attributes the success of one component to another.

E. Is target-language prosody predictable?

Table II gives the test-split result. Pitch level is not predictable beyond the baseline: the improvement is -0.02% and -0.00% , and the regulariser selected $\lambda = 10^6$, at which every weight vanishes, so the model reduced itself to B2. Pitch range is predictable in both directions, by 5.14% and 3.89% mean absolute error, with speaker-clustered p of 0.0012 and 0.0001 and bootstrap intervals excluding zero. The duration ratio is not reliably predictable.

TABLE II
PROSODY PREDICTION ON THE HELD-OUT TEST SPLIT (MEAN ABSOLUTE ERROR; LOWER IS BETTER). p IS A WILCOXON TEST ON PER-SPEAKER MEAN ABSOLUTE ERRORS, $n = 14$ SPEAKERS.

Dir.	Target	B1	B0	B2	Ridge	$\Delta B2$ (p)
EN→ES	F0 level	5.783	1.375	1.353	1.353	-0.02% (.0040)
EN→ES	F0 range	1.034	0.977	0.983	0.932	-5.14% (.0012)
EN→ES	Dur. ratio	0.181	0.186	0.181	0.180	-0.73% (.194)
ES→EN	F0 level	5.947	1.350	1.329	1.329	-0.00% (.903)
ES→EN	F0 range	1.017	0.963	0.968	0.931	-3.89% (.0001)
ES→EN	Dur. ratio	0.182	0.187	0.182	0.172	-5.34% (.104)

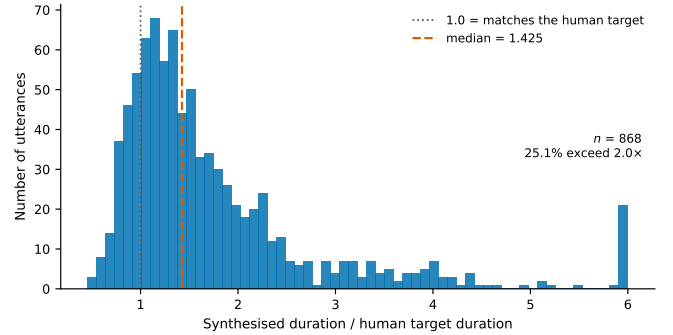


Fig. 5. Ratio of synthesised to human-target duration over 868 utterances, clipped at $6\times$ for display.

Under Bonferroni at $\alpha = 0.05$ the threshold is 0.0083, and three of six p -values fall below it; the same three survive Holm and Benjamini–Hochberg. Only two are practically meaningful, the pitch-range results. The third, EN→ES pitch level, has an effect of -0.02% : a signed-rank test over near-identical error vectors can return a small p for a zero-sized effect, and we report it so that it cannot be quoted as a third success.

The mechanism is in the coefficients. The weight on source F0 range is -0.191 (EN→ES) and -0.143 (ES→EN), large relative to the other five and *negative*: an unusually wide source pitch range should be predicted narrower in the target. Per-utterance variation regresses towards the speaker’s habitual range rather than transferring one for one. A conditioning mechanism that copied source pitch directly into the target would copy the wrong thing.

F. Duration inflation

Synthesised utterances run $1.52\times$ longer than the human target on average (4.07s against 2.67s); the median per-utterance ratio is 1.43 and 25.1% exceed $2.0\times$ (Fig. 5). The inflation is systematic, not random, and it is fatal for dubbing: at $1.5\times$ per utterance a dubbed track loses synchronisation within a few turns.

We built a video dubbing application that time-scales each synthesised segment to the slot it replaces using WORLD analysis and resynthesis [26], with scaling clamped to 0.5– $2.0\times$. Over five jobs and 83 segments, the dubbed track lands within 7% of the original length where the raw length ratio is at or below about $1.6\times$; at $2.17\times$ the clamp binds on 59% of segments and the output degrades. No amount of signal

processing absorbs a $2.17\times$ ratio: the constraint is upstream, in a translation whose spoken length was never controlled, which is exactly the problem length-aware translation for dubbing addresses [27], [28].

V. DISCUSSION

A. A measurement pitfall, quantified

An earlier version of this analysis reported that the system transferred essentially no pitch information, $r = 0.090$. Two defects produced that figure. Correlating pitch in hertz lets values furthest from the mean dominate, and a single octave-doubled observation destroys the statistic; correlating in semitones on identical measurements moved EN→ES F0 from 0.048 to 0.368. Adding the plausibility gate of Section III-D moved the same figure from 0.368 to 0.820.

A third intervention, attempted first, made matters worse. Narrowing the estimator’s *search* band to 60–400 Hz appears to be the natural remedy for octave errors; measured, it was the worst of four bands (generated-versus-human F0 correlation, semitone space, gated: 0.371/0.589 for 60–400 Hz; 0.778/0.739 for 65–600; 0.888/0.856 for 65–1000; 0.822/0.819 for 65–2093). The cause is that a narrow band starves the voiced/unvoiced decision of candidates: one test file went from ten voiced frames to zero. The correct design separates a wide search band from a narrow downstream plausibility gate. We report this because the erroneous number was the more publishable one, and because the plausible fix was the wrong fix.

B. Interpretation

The two headline results diverge because of the architecture, not because of an implementation defect. The speaker embedding on which zero-shot cloning depends is trained for verification and is designed to be invariant to what a speaker does on any particular utterance. A system built on such an embedding will reproduce *who* is speaking and discard *how* they spoke. Closing that gap requires an additional conditioning path, which is what expressive dubbing systems build [12], [13].

The prediction experiment qualifies this constructively. Cross-lingual prosody is not arbitrary, but its predictable component is narrower than one might hope, and compressive rather than one-for-one.

C. Limitations

No listening study was conducted; all results are instrumental, and whether listeners perceive the prosody gap as strongly as the correlations imply is untested. The study covers one corpus, one language pair and 14 test speakers, which is why speaker-clustered inference is used throughout. The corpus’s target side is a re-enactment rather than a literal translation, so translation scores against it are conservative. Prosody is characterised by utterance-level statistics, which do not capture contour shape; a frame-level comparison under dynamic time warping would be a stronger instrument and would make these figures directly comparable with published expressive-dubbing

scores. Finally, the prosody model is analysis, not control: it predicts target-language prosody but does not condition the synthesiser.

VI. CONCLUSION

A same-speaker parallel corpus converts cross-lingual synthesis evaluation from a set of uncalibrated similarity scores into a comparison against what the same person actually did. Under that comparison the answer is specific: current zero-shot cross-lingual voice cloning carries the speaker but not the speech act. Identity survives at 86–95 % of the human cross-language ceiling; within-speaker pitch level survives at roughly a third of the human figure and pitch range at essentially none. The predictable part of cross-lingual prosody lies in pitch range, and the relationship is compressive. Future work should validate these instrumental findings with a listening study, replace utterance-level statistics with warped frame-level contour comparison, and close the loop by using the prosody prediction to condition synthesis.

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